

Generative AI Curriculum



Generative AI

Course Duration: 3 months

Course Type: Certification course

Pre- requires: Laptop and Basic Python

Course Overview:

- Python
- LLMs (Large language models)
- RAG (Retrieval-Augmented Generation)
- Embedding
- Vector Store
- Prompt Engineering
- LangChain
- Llama Index
- Crew AI
- Guardrails and Masking PII (Personally Identified Information)
- Chatbot

Course Breakdown:

- Python and VS Code complete setup

- Virtual environment and jupyter notebook
- Github and Repo Intro
- Github copilot and other extensions
- **Basic Python**
 1. For loop
 2. While loop
 3. If, Ifelse statements
- Creating and calling Class and Functions
- Package building
- Generative AI — Basic theory, companies and LLMs
- Get LLM credentials (API key) from Gemini, OpenAI, HF, Azure Openai
- Basic LLM calls with temperature, top p, top k and max tokens.

What are tokens?

- LangChain — Basic theory and LLM calls
- Agentic Workflow with tools and agents with example
- **Prompt Engineering**
 1. Prompt techniques
 2. Prompt templates
- Embedding — Basic theory, getting credentials
- **RAG**
 1. Document loading
 2. Chunking strategies

- Splitters
 - Chunk size
 - Chunk overlap
 - Chunk size and token size
- **Vector Store (In-memory & Persistent)**
 - 1. ChromaDB
 - 2. FAISS
 - 3. Pinecone
 - 4. Azure AI search
- Agentic workflow with Prompt engineering, embedding, RAG and vector store
- Guardrails — Basics
- **Masking PII**
 - 1. HF
 - 2. Piiscan
 - 3. Azure Language service
- Llama Index (same as all from 12-19)
- Unstructured
- CrewAI — Basics with examples
- **ChatBot**
 - 1. Streamlit
 - 2. Chainlit